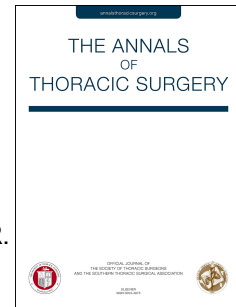


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**Bronchoscopy Simulation Training as a Tool in Medical School Education**

Running Title: Bronchoscopy Simulation in Med School

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**Abstract**

**Background:** Procedural simulation training is rare at the medical school level and little is known about its usefulness in improving anatomical understanding and procedural confidence in students. Our aim is to assess the impact of bronchoscopy simulation training on bronchial anatomy knowledge and technical skills in medical students.

**Methods:** Medical students were recruited by email, consented, and asked to fill out a survey regarding their baseline experience. Two thoracic surgeons measured their knowledge of bronchoscopy on a virtual reality bronchoscopy simulator using the Bronchoscopy Skills and Tasks Assessment Tool (BSTAT), a validated 65-point checklist (46 for anatomy, 19 for simulation). Students performed four self-directed training sessions of 15 minutes per week. A post-training survey and BSTAT were completed afterwards. Differences between pre- and post-training scores were analyzed with paired t-tests and random intercept linear regression models accounting for baseline BSTAT score, total training time, and training year.

**Results:** The study was completed by 47 medical students with a mean training time of  $81.5 \pm 26.8$  minutes. Mean total BSTAT score increased significantly from  $12.3 \pm 5.9$  to  $48.0 \pm 12.9$  ( $p < 0.0001$ ); mean scores for bronchial anatomy increased from  $0.1 \pm 0.9$  to  $31.1 \pm 12.3$  ( $p < 0.0001$ ) and bronchoscopy navigational skills increased from  $12.1 \pm 5.7$  to  $17.4 \pm 2.5$  ( $p < 0.0001$ ). Total training time and frequency of training did not significantly impact level of improvement.

**Conclusions:** Self-driven bronchoscopy simulation training in medical students led to improvements in bronchial anatomy knowledge and bronchoscopy skills. Further investigation is underway to determine the impact of bronchoscopy simulation training on future specialty interest and long-term skills retention.

Word Count: 250

Flexible bronchoscopy is an important procedure performed in various medical specialties such as thoracic and trauma surgery, pulmonology, critical care medicine, anesthesia, and otolaryngology. Historically, bronchoscopy education for physicians-in-training has been directly performed on patients. This method of skills acquisition, while adhering to the “see one, do one, teach one” apprenticeship model of medical education, poses risk for both the learner and the patient: procedural anxiety, inappropriate diagnostic or therapeutic results, post-procedure comorbidities, and reduced learning opportunities in low-caseload practices.

The first reported animal-model bronchoscopy skills exercise was reported by Ram et al. in the early 1990’s as otolaryngology residents practiced on porcine airways [1]. With the era of computerization, online bronchoscopy training has become a widely accessible and free learning resource, yet the user can only navigate the airway by clicking on bronchial segments and identifying anatomical landmarks. Virtual reality (VR) simulation on the other hand, teaches trainees flexible fiberoptic bronchoscopy handling skills while providing learners with added features such as haptic feedback during airway exploration. Other advanced features such as patient cough, use of lidocaine aliquots, timed procedural duration, recorded number of airway trauma, multiple patient airway scenarios, and biopsy options are also available on VR simulation models for advanced learning [2]. Other training platforms have also become accessible as creative clinicians find cost-effective methods to practice bronchoscopy skills by building paper maché airway models or 3D-printed bronchial trees in limited-resource environments [3, 4].

Support for the use of VR simulation training in bronchoscopy skill acquisition is widely found throughout the medical literature. When scrutinizing procedural outcomes, Blum et al. found that simulation-trained surgical residents experienced a significant decrease in both procedural times as well as visual and physical assistance needed, while competency was equivalent to that of traditionally trained residents [5]. Alternatively, Ost et al. demonstrated additional benefits of bronchoscopy simulation training such as increased procedure speed, higher percentage of lung segments visualized, and decreased airway trauma in the simulation group. From a patient safety perspective, practice on a simulator vs. live-

patients is associated with significantly lower amounts of patient discomfort and morbidity. Subjectively, participation in simulation-based training was associated with increased physician interest and satisfaction, while providing a realistic experience, even in expert-level bronchoscopy users [6]. In a 2015 CHEST expert panel review, Ernst et al. developed guidelines from evidence-based research and expert panelist opinions specifically referring to the importance of the use of simulation in bronchoscopy competency training [7].

The evidence supporting the use of bronchoscopy simulation training has been well validated in pulmonary fellows and residents, but little is known about the impact of bronchoscopy simulation on medical students' education. Early exposure to surgical skills education such as bronchoscopy simulation in undergraduate medical education can positively impact students' attitudes towards surgical careers and is hypothesized to translate into improved procedural skills in intern year of residency [8, 9]. As such, the aims of this study are to assess the impact of bronchoscopy simulation training on procedural skills and airway anatomy knowledge in medical students, and to investigate participating students' subjective attitudes towards VR simulation training and the quality of the self-directed training curriculum itself.

## Material and Methods

All full-time medical students at large private allopathic medical school were recruited to the study via email communication. Those students showing interest in participation were invited to attend an in-person information session describing the study aims and methodology. Informed consent was obtained by qualified research staff at the conclusion of the information session. Institutional Review Board approval was granted for the study.

A pre-training survey was administered to the participants prior to their first day of simulation training to collect information on baseline exposure to bronchoscopy and medical school (MS) year. Participants were asked to disclose prior experience observing or performing bronchoscopies. Next, study participants were given a brief introductory training on the use of the bronchoscopy simulator's basic operations and instrument handling by a simulation laboratory employee. The EndoVR Endoscopy

Simulator (CAE Healthcare, Canada), a high fidelity, haptic feedback virtual reality simulator was used for the purpose of this study (Figure 1) [10]. For their baseline skills examination (pre-training test), participants were evaluated on the simulator by one of two attending thoracic surgeons using a modified version of the validated Bronchoscopy Skills and Tasks Assessment Tool (BSTAT) evaluation tool (Figure 2) [11]. Participants were instructed to individually train on the simulator for a minimum of 15 minutes per week for a total of four consecutive weeks, following which participants were evaluated on the simulator for a second time (post-training test). The same thoracic surgeons evaluated participants with the modified BSTAT evaluation tool. A post-training survey was also administered to the participants at that time, assessing impressions about the quality and usefulness of the training, as well as students' subjective feeling of preparedness for future live bronchoscopies. Simulation metrics were retrieved from the simulator to measure each participant's total simulation training time and frequency.

Educational resources were made available to students for the self-study portion of the training, including the simulator's integrated "Introduction to Bronchoscopy" material which included video demonstrations, a virtual 3D model of airway anatomy, and text-guided bronchoscopy instructions. An annotated laminated diagram of the lung anatomy was also kept on the simulator console for rapid visual reference during training navigation. Recommended online educational resources included the "Bronchoscopy Step-by-Step®", a series of exercises designed by a consortium of international bronchoscopy experts teaching a systematic approach to bronchoscopy, and the [thoracicanesthesia.com](http://thoracicanesthesia.com) online bronchoscopy simulator [12, 13].

While the original BSTAT form is a validated scorecard used to evaluate trainee's bronchoscopy skills on a 100-point scale and includes tasks such as proper naming of pulmonary secretion and lung mucosa descriptions, as well as performance of bronchoalveolar lavage, mucosal biopsies and brushing on the bronchoscopy simulator [11], a truncated version of the BSTAT was used for the purpose of our study. Focusing mainly on airway anatomy and basic technical skills, the modified BSTAT used for evaluation purposes in this study was broken down into two main categories with multiple sub-tasks: airway anatomy knowledge (identification of left and right lung anatomy by segment during navigation,

identification of three respective left and right lung segments on demand), and bronchoscopy navigational skills (posture, hand position, and equipment safety, centeredness of scope in airway lumen, and avoidance of airway wall trauma). A maximum of 65 points could be scored on the modified BSTAT evaluation tool, composed of 46 points for the anatomy knowledge section, and 19 points for the bronchoscopy navigation skills portion of the BSTAT.

In order to quantitatively evaluate study participants' cognitive knowledge and technical skills acquisition, the measure of *class-average normalized gain* ( $\langle g \rangle$ ) was used. Already adopted as an evaluation metric in various academic fields such as mathematics, engineering, and biology, the class-average normalized gain measures a group of students' combined knowledge gain over the maximum possible improvement achievable. A class-average normalized gain of 30% or higher is considered to reflect curricular effectiveness [14]. By focusing on actual knowledge gain, the use of the class-average normalized gain helps both educators and researchers account for pre-teaching exposure to material:

$$\langle g \rangle = \frac{(\text{post-training class average}) - (\text{pre-training class average})}{[100\% - (\text{pre-training class average})]} \quad [14]$$

Summary measures (means, medians, standard deviations, proportions) were obtained for all variables. *Absolute gain* ( $\text{Mean Post-training scores} - \text{Mean Pre-training scores}$ )/( $\text{total possible score}$ ), *Relative gain* ( $\text{Absolute gain}/\text{Mean Pre-test score}$ ),  $\langle g \rangle$  "normalized gain", and  $g(\text{ave})$  were computed. Because post-test scores could potentially be lower than pretest (negative normalized gain),  $g(\text{ave})$  was included in the results. This essentially averages all  $\langle g \rangle$ , after replacing all negative gains with a 'zero' and deleting all of the respective negative students. In separate educational research, it has been shown that  $\langle g \rangle$  and  $g(\text{ave})$  are within 5% of each other as long as  $N > 20$ . [15]

Differences in mean pre- and post-training scores were tested with paired t-tests and random intercept linear regression models accounting for baseline BSTAT total score, total training time (or frequency of training) and training year. These analyses were also repeated for anatomical and simulation proficiency scores. All analysis was conducted in SAS 9.4 (SAS Institute Inc., Cary, NC) on students who completed the pre- and post-test. A two-sided  $p < 0.05$  was considered significant.

## Results

Out of 64 consented medical students, 47 completed the four weeks of bronchoscopy training (73.4%). While medical students from all classes were recruited, more than half of the participants who completed the training were first year medical students (53.2%). Twenty-five first year (53.2%), 16 second year (34.0%), one third year (2.1%), and five fourth year medical students (10.6%) completed the study. Students trained on the simulator for a mean time of  $81.5 \text{ min} \pm 26.8$ , with seven students completing three training sessions (15.2%), 28 students completing the required four training sessions (60.9%), and 11 students completing more than four sessions (23.9%), with a maximum of eight training sessions per participant.

Simulation testing scores, learning gains, and individual changes in pre- and post-training scores are shown in Table 1, Figure 3 and Figure 4 respectively. Mean BSTAT scores increased significantly from 18.9% ( $12.3/65 \pm \text{Standard Deviation (SD): } 5.9$ ) at pre-training to 73.8% ( $48.0/65 \pm 12.9$ ) at post-training ( $p < 0.0001$ ), with an absolute gain of 54.9% and a relative gain of 290%. Mean scores reflecting the students' knowledge of bronchial anatomy increased significantly from 0.2% ( $0.1/46 \pm 0.9$ ) to 67.6% ( $31.1/46 \pm 12.3$ ) ( $p < 0.0001$ ) following training, with an absolute gain of 67.4% and a relative gain of >3000%. Scores measuring participants' bronchoscopy navigational skills increased from 63.7% ( $12.1/19 \pm 5.7$ ) to 91.6% ( $17.4/19 \pm 2.5$ ) ( $p < 0.0001$ ), reflecting an absolute gain of 27.9% and a relative gain of 43.8%. The class-average normalized gain for the BSTAT exam was 67.7%, (67.6% and 79.8% for bronchial anatomy knowledge and bronchoscopy navigational skills, respectively). Accounting for total training time and MS year did not change the significant increase in the scores. While total training time and frequency were not associated with changes in BSTAT scores, first year medical students had a higher average BSTAT score than other medical students (32.3 vs. 28.1;  $p = 0.03$ ).

Following the completion of the study, 34 of 47 students (72.3%) reported finding the amount of simulation training time adequate, while 13 students (27.7%) wished more time was allotted for training, requesting an average additional time of  $64.6 \text{ minutes} \pm 36.9$ . No student considered the ascribed training



time to be excessive. Regarding the students' subjective rating of the training (Figure 5), 46 out of 47 students (97.9%) found the bronchoscopy simulation training useful with one student (2.1%) reporting a neutral view on the usefulness of the training. Thirty-six out of 47 students (76.6%) reported feeling well prepared for future live bronchoscopies, while six students (12.8%) reported not feeling well prepared for live bronchoscopies. Twenty-six students (55.3%) felt they could improve their bronchoscopy skills with additional simulation training, while eight participants (17%) did not believe additional training would improve their bronchoscopy skills. A total of 45 students (95.7%) indicated an interest in continuing bronchoscopy simulation training beyond the allotted training period. At the closure of the study, 20 students (42.6%) rated the quality of the self-directed bronchoscopy simulation training program as "excellent", 26 (55.3%) as "good", and one student (2.1%) rated the training program as "fair". In open-text comments, students suggested additional one-on-one training time with a thoracic surgeon, easier access to the simulation lab with extended training hours, as well as a real-life 3D visual model of the lung to further improve bronchoscopy proficiency.

### Comment

Thus far, medical literature on bronchoscopy simulation training has focused on resident-level providers and fellows. This study is the first English-language study to specifically investigate the use of virtual reality bronchoscopy simulation training in medical students. We found statistically significant improvement in both anatomical knowledge and navigational skills in our medical students following training, with a class-average normalized gain ( $\langle g \rangle$ ) of 67.9%, demonstrating the effectiveness of our month-long, self-directed bronchoscopy simulation curriculum. Furthermore, medical students' responses to a post-training survey revealed an overall positive experience that instilled procedural confidence and inspired continued interest in pursuing additional training for many participants.

Our study demonstrated an improvement in both airway anatomy knowledge and bronchoscope navigational skills in medical students. These results suggest that integration of bronchoscopy simulation training in undergraduate medical education may be a useful learning aid in courses such as anatomy or in

classes focusing on the respiratory system. During clinical years, bronchoscopy simulation training could support medical student's involvement in direct patient care and help learners develop a more concrete understanding of diagnostic and therapeutic interventions such as bronchial lavage, endobronchial ultrasound (EBUS), and other thoracic procedures. Furthermore, for those students entering fields with substantial bronchoscopy use, learning proper scope handling technique prior to starting residency could result in improved procedural confidence and reduced risk to patient in post-graduate medical years [9]. Ultimately, the role of virtual reality simulation training in novice learners is not well understood, and unanswered questions remain regarding the translation of simulation skills to real-life clinical outcomes.

First year medical students demonstrated a significantly higher average BSTAT score compared to their more senior schoolmates. While no clear factor can explain this phenomenon, it is possible that more recent or concurrent exposure to first-year curriculum anatomy class may provide MS1 students with a relative advantage compared to their colleagues. In addition, with MS2 students preoccupied with board exam preparation, and MS3/4 students rotating through clerkships of various difficulties, the relative amount of free time MS1 students enjoy compared to their colleagues may have contributed to increases off-simulator study time and subsequently improved BSTAT scores. Finally, as first year medical students represented 53.2% of the study's cohort, lower participation from other classes may have masked class-specific effects on score improvement.

Subjective impressions of the self-direct bronchoscopy simulation training were collected via a post-training survey in our participants (Figure 4). A majority of the students (76.6%) reported feeling well-prepared to perform live-bronchoscopies following completion of the study, demonstrating a confidence that is uncommon in the average medical student or even junior resident. When gauging how much interest medical students had in continuing the use of the bronchoscopy simulator following the conclusion of their training, 95.7% of the students felt that they would value further simulation training outside of the study, with 55.3% of participants stating they believed their bronchoscopy skills would continue improving with additional simulation training. As shown in previous studies, integration of hands-on clinical technologies, such as chest ultrasound when learning about respiratory physiology, can

lead to greater satisfaction, and improved knowledge-based test scores compared with non-exposed trainees among participating medical students [16]. As recent evidence reveals that millennial medical students value curricular flexibility and controlling the pace of their learning experiences [17], it is likely that the self-directed bronchoscopy simulation curriculum fit the modern pupil's learning style, contributing to the overall positive experience reported by our study's participants. These findings ultimately lend support to the utility of bronchoscopy simulation as a pedagogic and student-approved resource in medical school education.

Students' open-comment feedback collected in the post-training survey provided suggestions for improving the study. The most frequent student request, reported in 5 of the 17 total suggestions, was to have more one-on-one guided simulation time with an expert bronchoscopist. While integration of an in-person teacher in the current curriculum may negate the intention of a self-guided simulation training, guided assistance in the initial phase of the training would ease students' anxiety and provide tailored advice for intangible measurements such as posture, ease of manipulation, and equipment setup. Furthermore, with the current climate of online-accessible recorded lectures and anecdotal diminishing classroom attendance rates in medical school students, the opportunity to interact with attending-level physicians on a one-on-one basis can help foster mentorship, reel in students for valuable face-to-face interactions, and create opportunities for students to discover and utilize non-traditional learning resources beyond the bronchoscopy simulator in the laboratory.

To this day, the extent of VR bronchoscopy simulation training for medical professionals is unclear. While multiple studies investigating the impact of bronchoscopy simulation training on procedural skills found that novice bronchoscopists benefitted from VR training, the exact duration of training needed to achieve proficiency is yet to be determined. The first barrier to assessing the utility of simulation training in medical professionals lies in the lack of a standardized definition for bronchoscopy competency. Does competency pertain to navigational skills and anatomical knowledge alone, or should a competent bronchoscopist also master foreign object retrieval and EBUS? In addition to the lack of consensus on proficiency definition, training times of published bronchoscopy simulation protocols have

ranged from a single one-hour module to 10 respective training sessions [2]. Regardless of the simulation training's duration, however, researchers demonstrated improvement in novice physician's bronchoscopy skills. Despite the lack of tangible data demonstrating the translation of bronchoscopy simulation skills into actual positive patient outcomes, academic training programs and professional organizations such as the American College of Chest Physicians continue to employ and recommend VR bronchoscopy training, suggesting a general support of bronchoscopy simulation training in physicians' education.

Our study had several limitations. The single-institution sample was small, with only 47 medical students completing the simulation training program. A larger, multi-institutional sample with more heterogeneity in students' medical education background may provide more meaningful results. The use of a true, simulation-naïve control group in future studies could also help identify the effects of simulation training in our medical student population. Additionally, our limited timeline of four weeks of training followed by immediate post-training test did not facilitate the assessment of long-term skill retention. Instituting a long-term follow-up program, with interval evaluations at three, six, and 12 months post-training may provide more insight into the true level of skill retention associated with simulation training. Another consideration is the evaluators themselves. Although the BSTAT is a standardized grading tool, interpretations may vary. This bias can be addressed by training our evaluators and conducting mock testing to ensure that grading is consistent between evaluators. Our study used a truncated version of the BSTAT evaluation scale, allowing evaluators to focus on the assessment of bronchial anatomy knowledge and bronchoscopy navigation skills, as other skills such as mucosal pathology and bronchial secretion identification were not deemed critical learning elements for our study's medical school-level trainees. While a precedent exists for the use of a shortened BSTAT in bronchoscopy simulation training evaluation [14], altering the validated version of the BSTAT inherently jeopardizes its validity as an evaluation tool. Lastly, time elapsed between participants' final training sessions and the post-training evaluation was inconsistent. As students training closer to their final evaluation date may have a time-related advantage on their post-training BSTAT score, it may be beneficial to standardize this time interval in order to eliminate confounding by a proximity effect. As pointed out by participants in open-

comment feedback, facilitating students' access to the simulation laboratory may improve the ease of training, especially considering that 15.6% were unable to complete the study's required four training sessions.

Future directions of this study should focus on the impact of bronchoscopy simulation training on participants' career choices, as well as long-term skill retention associated with medical-student level simulation training. Administration of a follow-up survey at the conclusion of participants' medical school education could help determine whether an early exposure to bronchoscopy simulation training inspired interest in pulmonary, critical care, or surgical career choices. Alternatively, such follow-up could reveal whether students chose to continue exploring VR simulation options throughout their medical school years or decided to conduct medical education research utilizing simulation technologies. Students should also be re-evaluated for skill retention over time. With first and second year medical students representing 87.2% of the study participants, evaluation of skill retention once students have entered their clinical years, specifically during their surgery rotation, would be of great interest.

In conclusion, this study demonstrated the value of integrating bronchoscopy simulation training into undergraduate medical education, both from a pedagogic and knowledge and skills-acquisition standpoint. Despite a brief, four-week self-guided simulation training program, students' knowledge of bronchial anatomy and bronchoscopy navigational skills improved significantly. With positive feedback pertaining to both the utility and quality of the simulation training, medical educators ought to consider integrating VR bronchoscopy simulation training into their medical students' curriculum. Allowing medical students access to VR bronchoscopy simulation training is not only useful in practice, but can also provide a novel approach to engaging students in non-traditional and effective educational methods.

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Table 1. Gain in Bronchoscopy Skills and Tasks Assessment Tool scores

| BSTAT category                                      | Pre-training          | Post-training          | p-value <sup>a</sup> | Absolute Gain <sup>b</sup> | Relative Gain <sup>c</sup> | <g> <sup>d</sup> | g(ave) <sup>e</sup> |
|-----------------------------------------------------|-----------------------|------------------------|----------------------|----------------------------|----------------------------|------------------|---------------------|
|                                                     | Mean ± SD             |                        |                      |                            | (%)                        |                  | Mean(%) ± SD        |
| <i>Bronchial anatomy knowledge</i>                  | 0.1 ± 0.9<br>(0.2%)   | 31.1 ± 12.3<br>(67.6%) | <0.0001              | 67.4                       | 31000                      | 67.5             | 67.6 ± 26.8         |
| <i>Bronchoscope navigational skills<sup>f</sup></i> | 12.1 ± 5.7<br>(62.7%) | 17.4 ± 2.5<br>(91.6%)  | <0.0001              | 27.9                       | 43.8                       | 76.8             | 79.8 ± 36.5         |
| <i>Total BSTAT score</i>                            | 12.3 ± 5.9<br>(18.9%) | 48.0 ± 12.9<br>(73.8%) | <0.0001              | 54.9                       | 290                        | 67.7             | 67.9 ± 24.1         |

BSTAT: Bronchoscopy Skills and Tasks Assessment Tool

SD: standard deviation

<sup>a</sup> p-values obtained from paired t-tests<sup>b</sup> Absolute gain = [(post-training class average) - (pre-training class average)]/(total possible score)<sup>c</sup> Relative gain = [(post-training class average) - (pre-training class average)]/(pre-training class average)<sup>d</sup>  $<g> = \frac{(\text{post-training class average}) - (\text{pre-training class average})}{[100\% - (\text{pre-training class average})]}$ <sup>e</sup>  $g(\text{ave}) = \frac{\sum_{i=1}^N (g_i)}{N}$ ; Class average normalized gain > 30% reflecting curricular effectiveness



<sup>f</sup> Of 47 students who completed the pre- and post-training, 15 were excluded because of negative scores or no absolute gain in scores

### Figure Legends

Figure 1. EndoVR simulator training screen.

Figure 2. Modified Bronchoscopy Skills and Tasks Assessment Tool evaluation checklist.

RB: Right bronchus; LB: Left Bronchus

Figure 3. Comparison of Pre- vs. Post-Training Bronchoscopy Skills and Tasks Assessment Tool scores.

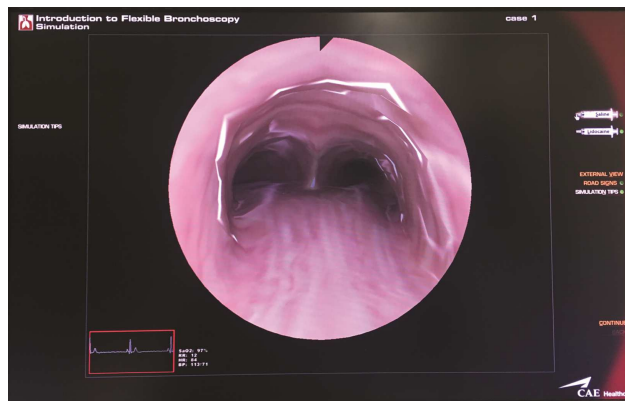
(a) BSTAT: Bronchoscopy Skills and Tasks Assessment Tool (b) Absolute gain = [(post-training class average) - (pre-training class average)] / (total possible score) (c) Relative gain = [(post-training class average) - (pre-training class average)] / (pre-training class average) (d) Average gain =  $\frac{\sum_{i=1}^N (g_i)}{N}$ ; Class

average normalized gain > 30% reflecting curricular effectiveness.

Figure 4. Pre- vs. Post-Training Total Bronchoscopy Skills and Tasks Assessment Tool scores.

BSTAT: Bronchoscopy Skills and Tasks Assessment Tool

Figure 5. Post-training survey results.



Modified Bronchoscopy Skills and Tasks Assessment Tool evaluation checklist

| <b>Educational Item</b><br>Items 1-10 each scored separately                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Satisfactory<br/>Yes/No</b> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| <b>1. Identification of Right sided anatomy (2 points each, target 20 points)</b><br><input type="checkbox"/> RB1 apical <input type="checkbox"/> RB2 posterior <input type="checkbox"/> RB3 anterior <input type="checkbox"/> RB4 lateral <input type="checkbox"/> RB5 medial<br><input type="checkbox"/> RB6 superior <input type="checkbox"/> RB7 mediobasal <input type="checkbox"/> RB8 anterobasal <input type="checkbox"/> RB9 laterobasal<br><input type="checkbox"/> RB10 posterobasal | Yes / No<br>Score ____/20      |
| <b>2. Identification of Left sided anatomy (2 points each, target 16 points)</b><br><input type="checkbox"/> LB1+2 apical/posterior <input type="checkbox"/> LB3 anterior <input type="checkbox"/> LB4 superior <input type="checkbox"/> LB5 inferior<br><input type="checkbox"/> LB6 superior <input type="checkbox"/> LB8 anterobasal <input type="checkbox"/> LB9 laterobasal <input type="checkbox"/> LB10 posterobasal                                                                     | Yes / No<br>Score ____/16      |
| <b>3. Identify and enter RB 4+5+6 on demand (All three segments must be entered to earn 5 points, no partial points given, target 5 points)</b><br><input type="checkbox"/> RB 4+5+6                                                                                                                                                                                                                                                                                                            | Yes / No<br>Score ____/5       |
| <b>4. Identify and enter LB 8+9+10 on demand (All three segments must be entered to earn 5 points, no partial points given, target 5 points)</b><br><input type="checkbox"/> LB 8+9+10                                                                                                                                                                                                                                                                                                          | Yes / No<br>Score ____/5       |
| <b>5. Posture/Hand positions/Equipment safety (3 points each, target 9 points)</b><br><input type="checkbox"/> Body posture <input type="checkbox"/> Hand positions <input type="checkbox"/> Equipment handling                                                                                                                                                                                                                                                                                 | Yes / No<br>Score ____/9       |
| <b>6. Scope centered and kept in midline (5 points, no partial points given)</b><br><input type="checkbox"/> Scope centered in airway lumen                                                                                                                                                                                                                                                                                                                                                     | Yes / No<br>Score ____/5       |
| <b>7. Airway wall trauma avoided (5 points, no partial points given)</b><br><input type="checkbox"/> Airway wall trauma avoided                                                                                                                                                                                                                                                                                                                                                                 | Yes / No<br>Score ____/5       |

